Binary Search Tree Iterator (medium)

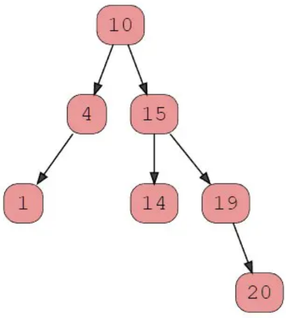
Problem Statement

Implement an iterator for the in-order traversal of a binary search tree (BST). That is, given a BST, we need to implement two functions:

bool hasNext(): Returns true if at least one element is left in the in-order traversal of the BST. **int next()**: Return the next element in the in-order traversal of the BST.

Example:

Given the following BST:



Here is the in-order traversal of this tree: [1, 4, 10, 14, 15, 19, 20]

Here is the expected output from the algorithm based on different calls:

hasNext() -> **true**

next() -> **1**

next() -> **4**

hasNext() -> **true**

next() -> **10**

next() -> **14**

next() -> **15**

next() -> **19**

next() -> **20**

hasNext() -> **false**

Try it yourself

Try solving this question here:

Java

```
import java.util.Stack;

/** definition for a binary search tree node */
/*class TreeNode {
  int val;
  TreeNode left;
  TreeNode right;
  public TreeNode(int val) {
    this.val = val;
    this.left = null;
    this.right = null;
  }
}*/

public class Solution {
  private Stack<TreeNode> stack = new Stack<TreeNode>();

  /** Write your class initialization code here */
  public void initialize(TreeNode root) {

  }
```

Test Case | Run Code Result | Stdout

[7,3,15,null,null,9,20]

Use Example Testcases | Run | Submit